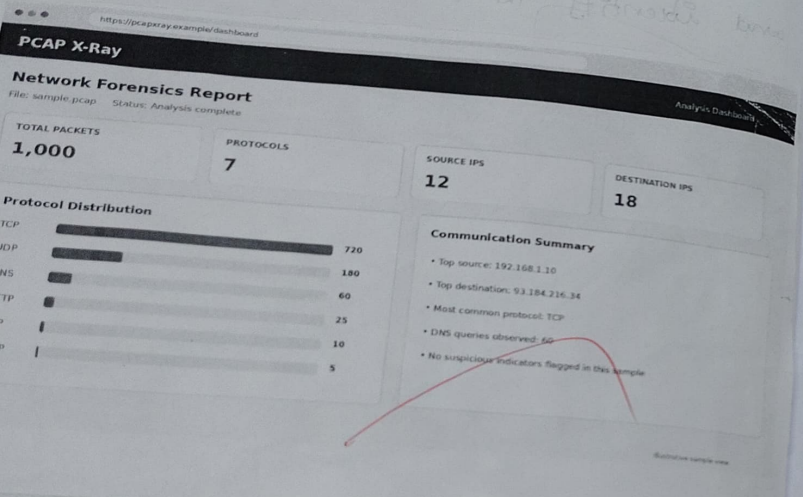


EXP 13.

Demonstrate network forensics using pcap X ray tools

AIM:

To analyze a pcap file and identify network protocols, communication pattern transferred files and suspicious activities

PROCEDURE:

* open a web browser and access the pcap x-ray tools
* obtain a sample PCAP files from a suitable source
* upload the PCAP file to the tool
* wait for the packet analysis to complete
* view the dashboard and note the total packets, protocols and IP address
* Examine the protocol distribution including TCP, UDP, DNS, HTTP and ARP
* Inspect hosts, conversation, transfer files and possible security indicators

https://pcapxray.example/dashboard

PCAP X-Ray

Analysis Dashboard

etwork Forensics Report

sample.pcap Status: Analysis complete

acket / Indicator Analysis

ne	Source	Destination	Protocol	Info
012	192.168.1.10	192.168.1.1	ARP	Who has 192.168.1.1?
48	192.168.1.10	8.8.8.8	DNS	Standard query A example.com
1	192.168.1.10	93.184.216.34	TCP	443 → 51527 [SYN, ACK]
	93.184.216.34	192.168.1.10	HTTPS	TLS application data
	192.168.1.10	8.8.8.8	ICMP	Echo request
	192.168.1.10	192.168.1.20	UDP	DNS response received

dicators: review unusual ports, suspicious destinations and large outbound transfers.

Illustrative sample view

- generate an downloadable analysis report if option is available

Result:

The PCAP file was analyzed using PCAP - X-Ray and network communication details and security indicator were examined